

Experiment 8: Face Detection using Haar Cascade Classifier

Theory:

Haar Cascade is a machine learning object detection algorithm used to identify objects in images or video. It is particularly known for face detection tasks. The algorithm uses Haar features, which are digital image features used in object recognition. A cascade function is trained from lots of positive and negative images, then it is used to detect objects in other images. The OpenCV library provides pre-trained Haar classifiers in XML format for frontal face detection. These classifiers scan the image at multiple scales and detect regions that match facial patterns. This experiment demonstrates how to use OpenCV's `CascadeClassifier` with a pre-trained frontal face Haar cascade model to detect and draw rectangles around detected faces in a static image.

Algorithm:

1. Load the Haar Cascade classifier using OpenCV's built-in frontal face XML file.
2. Read the input image using `cv2.imread()`.
3. Convert the image to grayscale using `cv2.cvtColor()`.
4. Use the `detectMultiScale()` method to detect faces in the grayscale image.
 - Parameters: `scaleFactor`, `minNeighbors`, `minSize` to adjust detection sensitivity.
5. Loop over the detected face coordinates and draw bounding rectangles using `cv2.rectangle()`.
6. Display the image using `cv2.imshow()` and hold the window open until a key is pressed.

Flowchart:

